

# SS-RESfM: Self-Supervised Outlier Handling Across Contamination Regimes in Deep Structure-from-Motion

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## Abstract

001 State-of-the-art deep equivariant Structure-from-  
002 Motion (RESfM) rejects outlier correspondences with  
003 a supervised classifier trained on COLMAP-derived  
004 labels. We present SS-RESfM, an entirely self-  
005 supervised alternative: the outlier head is trained on  
006 confident pseudo-labels formed from percentiles of the  
007 model’s own reprojection error, and no ground-truth or  
008 COLMAP labels are used anywhere in training or test.  
009 Under a strictly controlled protocol (identical tracks,  
010 robust bundle adjustment, and training schedule, vary-  
011 ing only the outlier mechanism) we compare label-free  
012 statistical removal (madweight) and soft reweighting  
013 against a from-scratch RESfM baseline across six  
014 datasets spanning 0.5%–61% measured outliers. The  
015 optimal mechanism flips with contamination. Removal  
016 wins the high-contamination regime: at 61% plain  
017 removal beats the supervised baseline on every seed  
018 ( $13.0 \pm 2.6^\circ$  vs.  $16.2^\circ$ ) and both beat COLMAP,  
019 while at 43% it nearly halves the supervised median  
020 on every seed ( $1.4\text{--}1.8^\circ$  vs.  $2.9^\circ$ ). Reweighting wins  
021 the low-contamination regime, beating the supervised  
022 baseline  $2.6\text{--}3.8\times$  on clean out-of-distribution data.  
023 Supervision retains the clean, in-distribution regime  
024 (self-supervised arms within  $\sim 15\text{--}20\%$ ). We trace  
025 this to a structural circularity: outlier labels require  
026 a trusted reconstruction of the very scenes whose  
027 contamination makes reconstruction hard (for any  
028 gold-standard labeler, not just COLMAP), and we  
029 measure the resulting label ceiling: regenerating  
030 supervision from scratch, label recall collapses from  
031 0.82 to  $\sim 0.3$  above  $\sim 15\%$  contamination, precisely  
032 where robustness is needed. Label-free adaptivity is  
033 thus a missing ingredient for robust deep SfM.

## 1. Introduction

Structure-from-Motion (SfM) recovers camera poses  
and 3D structure from unordered image collections.  
Its correspondences are heavily contaminated by out-  
liers (wrong matches that survive pairwise geometric  
verification), particularly in Internet photo collections  
where repeated and symmetric structure produces spu-  
rious epipolar geometries. Classical pipelines reject  
outliers with RANSAC and robust bundle adjustment  
(BA); recent deep equivariant methods (ESfM [2],  
RESfM [1]) instead learn a sets-of-sets permutation-  
equivariant network that regresses cameras from point  
tracks and, in RESfM, adds a supervised per-point in-  
lier/outlier classification module. RESfM’s labels are  
derived from a COLMAP [4] reconstruction in two  
stages: first by track membership, then refined by a  
4-pixel reprojection test against COLMAP-pose trian-  
gulation [1].

We take a different route. Rather than supervise  
the head with COLMAP labels, SS-RESfM trains it  
self-supervised: an adaptive confidence-weighted loss  
forms confident pseudo-labels from percentiles of the  
model’s own reprojection error, so no ground-truth or  
COLMAP outlier labels are used anywhere in train-  
ing or test. On top of this self-supervised head we  
study a question RESfM’s single supervised mecha-  
nism leaves open: is one outlier-handling mechanism  
optimal across the range of contamination levels SfM  
encounters? We find that it is not. Holding everything  
else fixed and varying only the outlier-handling mecha-  
nism, the best mechanism flips with contamination:  
hard removal at high contamination, soft reweighting  
at low, with the supervised classifier retaining the typ-  
ical in-distribution scene and the cleanest OOD mean  
(Fig. 1). We further show the removal family wins  
the high-contamination regime outright: plain removal  
beats the supervised baseline on every seed at both 43%  
and  $\sim 60\%$ , while madweight, which discards points  
by the median-absolute-deviation (MAD) of their re-

projection error and soft-weights the survivors with no learned threshold, matches the baseline there and beats COLMAP. We attribute this to (i) domain shift, since the classifier is trained only on in-distribution MegaDepth, and (ii) supervised labeling’s dependence on a COLMAP reconstruction and ground-truth poses, readily obtained for curated data but not for the contaminated collections where robustness is most needed; a label-free, per-scene mechanism is attractive precisely in that regime.

The appeal of self-supervision here is structural, not incidental. Outlier labels must be derived from a trusted reconstruction of the very scenes whose contamination makes reconstruction hard, so supervision bootstraps from precisely the capability it is meant to provide. This circularity is not specific to COLMAP: any gold-standard labeler must first solve the scene in order to label its outliers, and it is exactly the challenging, contaminated collections that defeat this bootstrapping. A self-supervised head, by contrast, consumes only the point tracks themselves; the training data for robust SfM then becomes as unlimited as unlabeled imagery, and new domains require no labeling infrastructure at all.

Contributions. (1) SS-RESfM, an entirely self-supervised outlier-handling approach for deep SfM: an adaptive confident-pseudo-label head (trained on percentiles of the model’s own reprojection error), label-free statistical removal, and robust BA, using no ground-truth or COLMAP outlier labels, in contrast to RESfM’s COLMAP-supervised head. It beats the supervised baseline on clean out-of-distribution data and at high contamination. (2) A controlled study of label-free removal vs. soft reweighting ( $\pm$ test-time fine-tuning) under an identical tracks + BA + schedule pipeline, isolating the mechanism alone. (3) The complementarity finding: the optimal mechanism flips with contamination; no single mechanism dominates. (4) madweight, which removes outliers by robust reprojection statistics (MAD), needing not even the head, and reweights survivors; it matches the from-scratch supervised baseline at extreme contamination ( $\sim 60\%$ ) with no learned threshold, while plain removal on the same self-supervised head beats the baseline outright, and both beat COLMAP, which registers only two-thirds of the cameras.

## 2. Related Work

Deep equivariant SfM. Learning-based SfM regresses cameras from point tracks with permutation-equivariant networks. ESfM [2] casts uncalibrated SfM as deep projective factorization over a measurement tensor, trained with an unsupervised reprojection

loss; GASfM [3] uses a graph-attention variant. Both assume outliers were already removed by two-view geometric verification. RESfM [1] targets contaminated internet collections with a sets-of-sets architecture plus a supervised per-point inlier/outlier classification module and a robust BA, reporting accuracy close to its own clean-track upper bound (“ESfM\*”) and on par with classical Theia/GLOMAP (often with better translation recovery, while registering fewer images). End-to-end pointmap and pose-regression pipelines (VGGSfM, MAST3R, DUSt3R) target far smaller image sets. We propose no new architecture; we reuse the RESfM/ESfM backbone and study the outlier-handling stage in isolation.

Outlier rejection. Correspondence outliers that survive RANSAC/two-view filtering are the central difficulty in internet-scale SfM. Classical pipelines suppress them with M-estimators and robust BA; RESfM casts rejection as supervised classification. We contrast supervised classification against label-free statistical removal via the median absolute deviation (MAD) [5] and soft reweighting, and show their relative merit is contamination-dependent.

Robust bundle adjustment. Following [1], we apply a robust BA that removes high-reprojection and low-visibility points, keeps the largest connected component, re-triangulates, and re-optimizes. This stage is identical across all our arms.

Test-time adaptation. Per-scene fine-tuning at inference is used by ESfM/ RESfM to reach sub-pixel accuracy. We treat it as an orthogonal option ( $\pm$ TTT) applied uniformly across mechanisms.

## 3. Method

### 3.1. Background

The input is a (sparse) point-track tensor  $M$  of  $m$  tracks across  $n$  cameras, built by concatenating pairwise RANSAC-verified SIFT matches. The sets-of-sets permutation-equivariant network [1, 2] maps  $M$  to per-camera poses and a per-point outlier score  $s \in [0, 1]$  (points in the same track may receive different scores). Crucially, unlike RESfM, whose classification head is trained with a BCE loss against COLMAP-derived inlier/outlier labels, SS-RESfM trains this head self-supervised, with an adaptive confidence-weighted loss: per forward pass it forms confident pseudo-labels from percentiles of the model’s own reprojection error (lowest  $p\%$  inliers, highest  $p\%$  outliers) and applies BCE only to those. No ground-truth or COLMAP outlier labels are used at any point. A reprojection error  $e$  is available for every observation once an initial reconstruction exists. All mechanisms below consume

$s$  (and, for madweight,  $e$ ), produce a per-observation weight  $w$  (or a hard keep/drop decision), and feed the result to the identical robust-BA stage (Fig. 2).

### 3.2. Outlier-handling mechanisms

We compare four mechanisms, illustrated as weight-vs-signal transfer functions in Fig. 2:

- remove (the RESfM analogue): drop a point if  $s > \tau$ ; keep with  $w=1$  otherwise. We use RESfM’s default  $\tau=0.6$  and its  $\tau=0.8$  for the low-outlier, single-camera Strecha/BlendedMVS sets [1].
- weight: soft,  $w = 1 - s$ , no removal.
- hybrid: three-band: keep ( $s$  below a low percentile), soft-weight (middle), drop ( $s$  above a high percentile).
- madweight (ours): removal is driven by robust reprojection statistics, not the head. An observation is removed if its reprojection error exceeds  $\text{med}(e) + \alpha \cdot \text{MAD}(e)$  ( $\alpha=2$ ); surviving observations are soft-weighted by  $w = 1 - s$ . madweight is the only mechanism that consumes the reprojection branch  $e$  and requires no learned threshold for removal.

Each mechanism is evaluated with and without test-time fine-tuning ( $\pm\text{TTT}$ ), in which the head continues to adapt on the test scene.

### 3.3. Test-time fine-tuning and robust BA

We load a multi-scene checkpoint, run a per-scene forward pass to obtain  $s$  and  $e$ , apply the mechanism, fine-tune per scene ( $\sim 1000$  iterations) under the reprojection objective, and apply robust BA. This protocol, the seed, and the schedule are held fixed across all arms; only the mechanism block changes.

## 4. Experiments

### 4.1. Datasets and contamination

We evaluate on six datasets spanning a wide contamination range. Alongside the RESfM paper’s quoted outlier fractions we report the contamination measured on our own tracks (robust ground-truth-pose triangulation with the standard 4px cutoff, applied uniformly to every dataset):

| Dataset        | Out% (RESfM) | Out% (ours) | Regime            |
|----------------|--------------|-------------|-------------------|
| Strecha [8]    | 2.7          | 1.7         | clean, LIDAR GT   |
| BlendedMVS [9] | 3.6          | 3.1         | low-outlier       |
| MegaDepth [7]  | 25.6         | 25.4        | contam., in-dist. |
| 1DSfM [6]      | 31.9         | 43.3        | contam., OOD      |
| 1DSfM-hard     | –            | $\sim 60$   | extreme, OOD      |
| Olsson [10]    | –            | 0.5         | outlier-free      |

On MegaDepth (where tracks are identical to RESfM’s) the two measurements agree (25.4 vs 25.6), validating our measurement; our 1DSfM rebuild is markedly harder than the tracks behind RESfM’s quoted 31.9%, which should be kept in mind when reading the 1DSfM columns. 1DSfM-hard comprises the five 1DSfM scenes unused by the standard ten-scene split (Gendarmenmarkt, Piccadilly, Roman Forum, Trafalgar, Union Square), built with the same pipeline and subsampled to 300 cameras per scene following RESfM’s convention. Networks are trained on 27 MegaDepth scenes; MegaDepth test is therefore in-distribution, while 1DSfM/1DSfM-hard/Strecha/BlendedMVS/Olsson are cross-dataset (out-of-distribution).

### 4.2. Protocol, baselines, and track provenance

Following RESfM’s public training code and default configuration, all networks train for 20,000 epochs with seed 20 (the seed reported by RESfM [1]) as the primary seed; the self-supervised arms are additionally replicated with seeds 21/22, and cells marked  $\dagger$  report mean $\pm$ std over the three seeds (the supervised baseline’s seed replicates were still training at submission, as noted per table). Batch size, image subsampling, the eleven-milestone learning-rate decay schedule (10k, 11k, ..., 20k,  $\gamma=0.9$ ), and validation-based early stopping follow that released configuration and are held identical across all arms. Two deliberate settings are disclosed. First, the base learning rate is per-method:  $10^{-3}$  for the supervised arms (RESfM’s released value) and  $10^{-4}$  for the self-supervised arms, whose adaptive pseudo-label loss converges early (best checkpoint by epoch  $\sim 7.5\text{k}$ , before the first decay milestone; the supervised arms peak at epoch  $\sim 16.5\text{k}$ ); both are well within the budget, so neither method is truncation-limited. Second, two secondary values (the infinity-point margin and the validation metric used for checkpoint selection) are held identical across all arms rather than at RESfM’s released values, preserving the controlled comparison. We report two architectures: shallow ( $1\times 3$ ) and deep ( $2\times 3$ ).

Fair-comparison anchor. Our controlled baseline is RESfM-from-scratch: trained by us with the identical setup as our arms (same 27 scenes, sampling, epochs, seed), so only the outlier mechanism differs. We additionally report RESfM-released (authors’ weights) and RESfM (paper).<sup>1</sup>

<sup>1</sup>The ESfM row likewise follows the unified 20k-epoch protocol. ESfM’s own released multi-scene (calibrated) recipe trains for 30k epochs plus a 500-epoch per-scene fine-tune; we deliberately hold the schedule fixed across all rows so that the outlier mechanism remains the only controlled variable.

Track provenance. MegaDepth tracks, preprocessing, and the 36 test scenes are identical to RESfM, so MegaDepth is directly comparable to published numbers. The OOD tracks (1DSfM/1DSfM-hard/Strecha/BlendedMVS/Olsson) are reconstructed by us (RESfM did not release OOD tracks); OOD comparisons are therefore controlled within our pipeline and are not directly comparable to RESfM’s published OOD numbers. On MegaDepth, RESfM’s own released weights yield  $0.209^\circ$  in our harness (vs. their reported  $0.169^\circ$ ), confirming our evaluation is faithful and that  $\sim 0.209^\circ$  is the reproducible level; this is unchanged ( $0.217^\circ$ ) under the authors’ exact pinned software stack (MKL BLAS, same pycolmap/pyceres), isolating the residual gap to solver/hardware-level nondeterminism in robust BA (analysis in the supplementary material). All reported results come from this single pipeline and software stack; a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism and should be read with that variance in mind.

Schedule robustness. All reported results use RESfM’s released eleven-milestone schedule. An earlier accidental five-milestone variant of ours was retired; a three-seed replication of both schedules shows the schedule effect lies within seed noise (MegaDepth  $0.46 \pm 0.08$  vs.  $0.40 \pm 0.09$ ; 1DSfM  $9.6 \pm 3.2$  vs.  $12.2 \pm 2.4$ , per-seed rankings inverting on both), with mechanism orderings stable throughout.

A method-isolated comparison, not a SOTA claim. Because RESfM-from-scratch shares every pipeline component with our arms and differs only in the outlier mechanism, it isolates the effect of removing supervision, the same controlled protocol RESfM itself uses when it re-trains ESfM/GASfM in its own pipeline. We do not claim state of the art: RESfM’s released checkpoint ( $0.209^\circ$  on MegaDepth) is stronger than either from-scratch arm. This reproduction gap stems from the released weights themselves, not our pipeline (the authors’ own training code on our tracks reproduces  $0.34\text{--}0.39^\circ$ , not  $0.209^\circ$ ), so it handicaps the supervised and self-supervised arms equally. We report RESfM-released and RESfM (paper) as reference rows throughout.

Metrics. Following RESfM’s convention we draw conclusions from mean errors (translation emphasized, rotation reported alongside), post-BA, in degrees, and provide medians as supporting statistics (RESfM likewise reports medians in its appendix). The one declared exception is 1DSfM, whose mean is dominated by two symmetric-structure failure scenes (Ellis Island, Tower of London) that fail for all methods; there we

treat the median as the representative figure.

## 5. Results

### 5.1. Main comparison

Table 1 is the controlled head-to-head under the RESfM-aligned schedule: every row shares tracks, architecture, schedule, robust BA, and seed, so any difference is attributable to the outlier mechanism and its supervision alone. Reading along the contamination axis: (1) Clean OOD (Olsson 0.5%, Strecha 1.7%, BlendedMVS 3.1%): the label-free arms win decisively where the supervised head misfires: every arm beats RESfM by  $2.6\text{--}3.8\times$  on BlendedMVS ( $0.10\text{--}0.15$  vs.  $0.38$ ) and by  $\sim 3\times$  on Olsson means ( $2.8 \pm 0.3$  over seeds 20–22 vs.  $8.5$ ; the supervised baseline collapses on a subset of the 39 scenes). Strecha is the exception: RESfM’s mean is far better ( $0.006$  vs.  $\sim 2$ , a single self-supervised failure scene), the medians tie ( $0.005$ ), and RESfM keeps the Olsson median ( $0.17$ ). (2) In distribution (MegaDepth), supervised RESfM retains mean and median ( $0.344/0.07$ ); the deterministic self-supervised arms land within  $\sim 23\%$  ( $0.42$ ) and the TTT arm averages  $0.39 \pm 0.07$  over evaluation repeats; near-parity without labels, not a win. (3) Moderate contamination (1DSfM, 43%) has no stable winner between madweight and the baseline ( $12.4/4.4$  vs.  $11.0/2.9$  at this seed; the retired five-milestone runs showed the reverse, and the three-seed replication brackets both), so we decline a directional claim for madweight. The removal family, however, is seed-consistent here: remove+TTT beats the baseline mean on every seed ( $8.7 \pm 1.1$  vs.  $11.0$ ) and plain remove attains median  $1.4\text{--}1.8$  vs.  $2.9$  across all three seeds (its means,  $9.1 \pm 2.8$ , remain failure-scene-dominated): a label-free win at 43% (Table 2), albeit against the baseline’s single available seed. (4) Extreme contamination (1DSfM-hard,  $\sim 60\%$ ): removal beats weighting robustly, madweight ( $17.0 \pm 4.4$  over seeds 20–22) statistically matches the supervised baseline ( $16.2$ , seed 20), and plain remove goes further:  $13.0 \pm 2.6$ , below the baseline on every seed, with tight medians ( $11.9 \pm 0.6$  vs.  $13.1$ ); all label-free arms are far ahead of COLMAP (§5.4). Rotation means (Fig. 1, right) follow the same ordering on MegaDepth/Strecha/BlendedMVS/Olsson (e.g. BlendedMVS  $6.6\text{--}9.6^\circ$  self-supervised vs.  $36^\circ$  supervised) and mildly favor the self-supervised arms on 1DSfM ( $8.3\text{--}8.4^\circ$  vs.  $9.8^\circ$ ) even where translation favors the baseline.

### 5.2. The complementarity finding

Fig. 1 shows Table 1 as a picture, and one pattern organizes it. Between supervision regimes, the self-

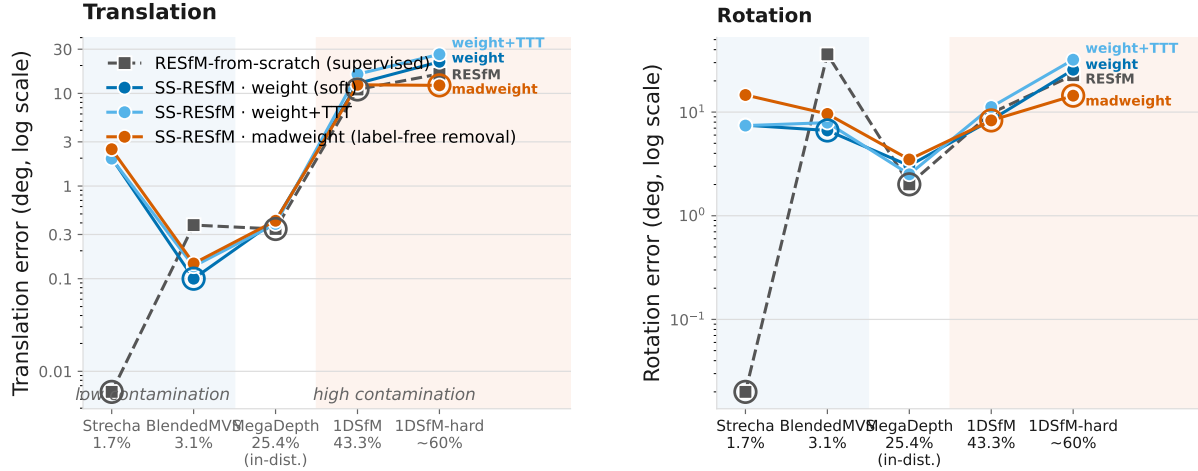


Figure 1. The optimal outlier mechanism flips with contamination. Translation (left) and rotation (right) means (log scale) of the self-supervised arms vs. supervised RESfM, five of the six datasets ordered by measured contamination (Table 1 data). Circled = best per dataset. The rotation panel mirrors the translation ordering, with the self-supervised arms also taking 1DSfM and the supervised BlendedMVS failure even more pronounced (36°).

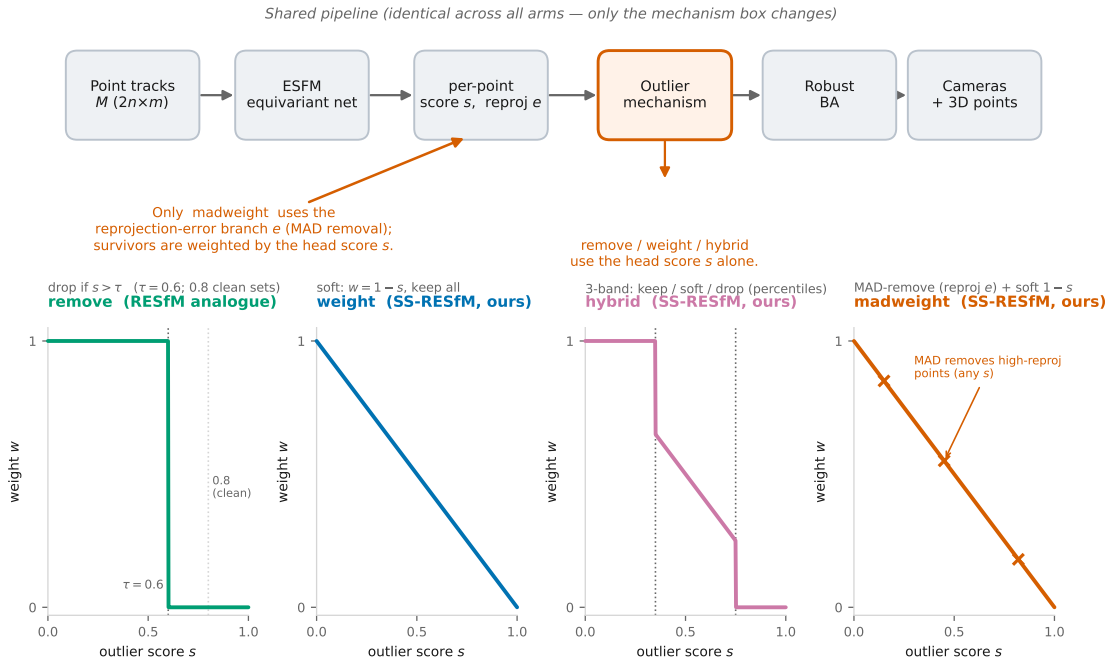


Figure 2. Method. A shared pipeline (top) in which only the outlier-mechanism block changes. The four mechanisms as weight-vs-signal transfer functions (bottom). Only madweight consumes the reprojection-error branch  $e$  for removal; the others use the head score  $s$ .

supervised wins concentrate exactly where a label-free design should help: on clean OOD data, where a classifier trained on contaminated MegaDepth misfires (on BlendedMVS, RESfM’s rotation error explodes to 36°), and at extreme contamination, where supervision itself can no longer be manufactured (§5.3). Within the self-

supervised family, the removal-vs-weighting ordering is monotone in contamination: weighting is best at 2–3%, the two tie in distribution, and removal dominates from 43% upward (madweight 12.4 vs. weight 12.8, plain remove 8.9, Table 2;  $13.0 \pm 2.6$  vs.  $\sim 22$  at 60%). This mechanism-level flip, the answer to our central ques-

Table 1. Main controlled comparison (RESfM-aligned schedule). Mean/median translation error (deg, post-BA), seed 20. All rows share tracks, architecture, schedule, robust BA, and seed; only the outlier mechanism and its supervision differ. Reference rows (means) below the rule; upper block = SS-RESfM arms. Best per column in bold. <sup>†</sup>mean $\pm$ std over three evaluation repeats (TTT is run-to-run nondeterministic; supplementary). <sup>‡</sup>mean $\pm$ std over seeds 20–22; the RESfM cell is seed 20 (its seed replicates were still training at submission). Other cells: deterministic, seed 20.

| Arm                        | MegaDepth (25.4)                  | 1DSfM (43.3) | 1DSfM-hard (60)                  | Strecha (1.7) | BMVS (3.1)  | Olsson (0.5)                   |
|----------------------------|-----------------------------------|--------------|----------------------------------|---------------|-------------|--------------------------------|
| madweight                  | 0.422/0.145                       | 12.39/4.37   | 17.0 $\pm$ 4.4/11.7 <sup>†</sup> | 2.50/0.97     | 0.146/0.004 | 2.8 $\pm$ 0.3/1.2 <sup>‡</sup> |
| weight                     | 0.423/0.182                       | 12.79/11.01  | 21.9 $\pm$ 3.4/12.7 <sup>†</sup> | 2.00/0.026    | 0.100/0.036 | 2.8 $\pm$ 0.4/1.1 <sup>‡</sup> |
| weight+TTT                 | 0.39 $\pm$ 0.07/0.14 <sup>†</sup> | 16.13/7.16   | 21.9 $\pm$ 3.9/14.5 <sup>†</sup> | 1.98/0.005    | 0.136/0.011 | 7.74/0.18                      |
| RESfM-scratch (supervised) | 0.344/0.070                       | 11.03/2.93   | 16.22/13.08                      | 0.006/0.005   | 0.379/0.348 | 8.47/0.17                      |
| RESfM (released)           | 0.209                             | 10.71        | –                                | 0.197         | 0.330       | –                              |
| RESfM (paper)              | 0.169                             | –            | –                                | –             | –           | –                              |

tion, is stable across schedules and seeds.

### 5.3. Why label-free removal wins under high contamination

Two factors explain why the supervised advantage erodes as contamination rises. First, domain shift: the classifier is trained only on in-distribution MegaDepth, and its decision boundary does not transfer to the outlier statistics of out-of-distribution internet collections (a domain-shift effect, not a labeling failure: the strong in-distribution result confirms the labels are sound). Second, and more fundamentally, supervision itself degrades in this regime: RESfM’s labeling (Appendix C of [1]) assumes a COLMAP reconstruction and ground-truth poses per training scene: readily obtained for curated data, but exactly what one lacks on the contaminated collections where robustness is most needed. madweight requires neither: it recomputes the outlier test per-scene from reprojection statistics, so it is unaffected by domain shift and applicable wherever a partial reconstruction exists. This is why the removal family prevails at extreme contamination (§5.4); madweight shares that robustness while requiring no learned threshold at all.

Measuring the label ceiling. We make the second argument quantitative. On 18 scenes spanning the full contamination axis we simulate the realistic labeling scenario for a new dataset (no ground-truth poses, so labels must come from a from-scratch COLMAP reconstruction of the same tracks with fixed known intrinsics) and score the resulting per-observation labels against the ground-truth-pose reference labels (precision/recall with outlier as the positive class; coverage is the fraction of observations that receive any COLMAP verdict at all, i.e. camera registered and track triangulated; scenes capped at 120 cameras):

| Contam. | $n$ | Cover% | Prec. | Recall | COLMAP $t$ -err |
|---------|-----|--------|-------|--------|-----------------|
| 0–5%    | 6   | 99.4   | 0.76  | 0.82   | 0.004           |
| 5–15%   | 4   | 89.6   | 0.90  | 0.83   | 0.014           |
| 15–30%  | 2   | 71.7   | 0.83  | 0.26   | 1.02            |
| 30–45%  | 3   | 55.7   | 0.74  | 0.39   | 0.82            |
| 45–60%  | 3   | 55.1   | 0.87  | 0.39   | 11.0            |

Precision stays high throughout, but recall collapses from 0.82 to 0.26–0.39 and coverage halves once contamination exceeds  $\sim 15\%$ : the corrupted reconstruction absorbs enough outliers that most reproject acceptably under its distorted geometry, and half the observations receive no label at all. A head trained on such labels is actively taught that most true outliers are inliers, precisely where robustness matters. Conversely, below  $\sim 15\%$  COLMAP-derived labels are excellent, consistent with RESfM’s strong in-distribution result and delimiting where the supervised recipe is sound.

### 5.4. Extreme contamination: the 1DSfM-hard set

To probe beyond 1DSfM’s 43%, we evaluate on the five held-out 1DSfM-hard scenes ( $\sim 60\%$  measured contamination, 300 cameras each). The learned arms are those of Table 1 (this restates its 1DSfM-hard column alongside rotation), and COLMAP runs incremental mapping on the same tracks with known intrinsics.

| Method        | Trans/Rot (post-BA) | Registered |
|---------------|---------------------|------------|
| remove        | 10.6/13.6           | all        |
| madweight     | 12.2/14.4           | all        |
| RESfM-scratch | 16.2/22.7           | all        |
| COLMAP        | 20.9/–              | 64%        |
| weight        | 21.9/25.6           | all        |
| weight+TTT    | 26.4/32.0           | all        |

The mechanism-level ordering extrapolates: over seeds 20–22, plain remove attains  $13.0 \pm 2.6$  translation (below the supervised 16.2, seed 20, on every seed; medians  $11.9 \pm 0.6$  vs. 13.1), madweight at  $17.0 \pm 4.4$  statistically matches it, the soft arms trail at  $\sim 22$ , and

Table 2. Full shallow mechanism matrix (RESfM-aligned schedule, seed 20;  $\ddagger$ mean $\pm$ std over seeds 20–22). Mean / median translation (deg, post-BA). Best per column in bold.

| Arm          | MD        | 1DSfM                                    | Strecha     | BMVS       |
|--------------|-----------|------------------------------------------|-------------|------------|
| remove       | 0.60/0.19 | 9.1 $\pm$ 2.8/1.6 $\pm$ 0.2 $\ddagger$   | 3.02/1.68   | 0.27/0.19  |
| remove+TTT   | 0.49/0.13 | 8.7 $\pm$ 1.1/2.2 $\pm$ 1.1 $\ddagger$   | 2.97/1.56   | 0.19/0.07  |
| hybrid       | 0.37/0.20 | 16.0 $\pm$ 1.4/6.7 $\pm$ 4.4 $\ddagger$  | 2.70/1.39   | 0.10/0.03  |
| hybrid+TTT   | 0.39/0.14 | 15.4 $\pm$ 1.1/10.7 $\pm$ 3.5 $\ddagger$ | 3.52/2.65   | 0.13/0.00  |
| madweight    | 0.42/0.15 | 11.5 $\pm$ 2.3/6.0 $\pm$ 1.4 $\ddagger$  | 2.50/0.97   | 0.15/0.004 |
| weight       | 0.42/0.18 | 15.8 $\pm$ 3.1/10.9 $\pm$ 2.1 $\ddagger$ | 2.00/0.03   | 0.10/0.04  |
| weight+TTT   | 0.39/0.14 | 17.0 $\pm$ 1.3/10.9 $\pm$ 4.1 $\ddagger$ | 1.98/0.005  | 0.14/0.01  |
| RESfM (sup.) | 0.34/0.07 | 11.03/2.93                               | 0.006/0.005 | 0.38/0.35  |

COLMAP degrades severely: its error is computed only over the 64% of cameras it registers (a favorable metric) yet still loses. Errors are large for every method (these scenes are hard), but the trend matches the label-ceiling prediction: the supervised advantage vanishes exactly where supervision can no longer be manufactured.

## 5.5. Full mechanism matrix

Table 2 completes the mechanism study (all arms, RESfM-aligned schedule, seed 20). Notably, at 43% the plain-removal family leads every arm including the supervised baseline, and the full three-seed band confirms it: remove+TTT 8.7 $\pm$ 1.1 mean (every seed below the baseline’s 11.0), plain remove median 1.59  $\pm$  0.18 (vs. 2.9; means 9.1  $\pm$  2.8, dominated by the two failure scenes). On clean data the soft arms dominate the removal family, reproducing the complementarity.

## 5.6. Which robust statistic? MAD vs. STD vs. Huber

madweight removes by a robust statistic of the reprojection error; we ablate the statistic holding the head reweighting fixed: MAD (median +  $\alpha \cdot 1.4826$  MAD, 50% breakdown), a Huber M-estimate of scale (IRLS), and non-robust STD (mean +  $\alpha\sigma$ ), all at  $\alpha=2$ . On contaminated 1DSfM, across seeds 20–22, MAD’s median is stable (4.4–7.0), Huber’s is volatile (3.8–12.4; better than STD on average but not dependably), and non-robust STD is consistently poor (9.0–13.3): its mean/ $\sigma$  threshold is inflated by the very outliers it should catch and under-removes. The 50%-breakdown statistic is thus the only reliably robust choice at high contamination. On clean Strecha/BlendedMVS all three are comparable (BlendedMVS means 0.11–0.15): little is removed either way, the right behavior without outliers. Robust removal is essential under contamination and superfluous on clean data, reinforcing the complementarity; we use MAD in madweight.

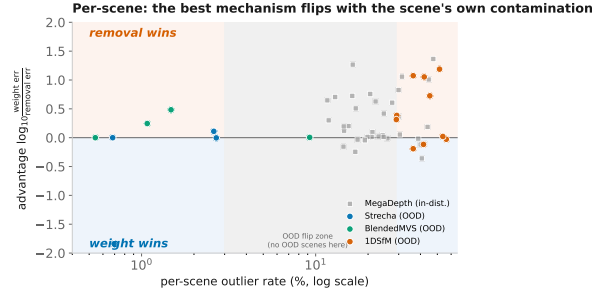


Figure 3. Per-scene complementarity (RESfM-aligned schedule). Each point is one scene: outlier rate (x, log) vs. signed advantage of removal over soft weighting. Weight wins at low contamination, removal at high contamination.

Table 3. Per-scene selection (mean translation, deg; RESfM-aligned schedule). Contamination-keyed selection collapses to per-dataset choice. The oracle uses ground-truth error over five mechanisms: an upper bound, not a deployable method.

| Dataset    | weight | madwt | contam | oracle (GT) |
|------------|--------|-------|--------|-------------|
| MegaDepth  | 0.423  | 0.422 | 0.422  | 0.116       |
| 1DSfM      | 12.79  | 12.39 | 12.39  | 5.78        |
| Strecha    | 2.00   | 2.50  | 2.00   | 1.98        |
| BlendedMVS | 0.100  | 0.146 | 0.100  | 0.083       |

## 5.7. Per-scene complementarity and the limits of contamination-based selection

Because individual scenes span a wide contamination range, we test whether the flip also holds per scene. Fig. 3 plots each scene’s outlier rate against the signed advantage of the best removal mechanism over soft weighting: on OOD data soft weight wins the low-contamination scenes and removal the high-contamination ones, with the crossover bracketed to the sparsely-sampled  $\sim 3$ –29% band.

Does per-scene contamination help selection? No: within each dataset contamination is too homogeneous to trigger different choices, so a contamination-keyed selector (madweight if scene rate  $\geq 10\%$ , else weight) collapses to per-dataset selection (Table 3). An oracle choosing the best of five mechanisms per scene from ground-truth error would change the picture: MegaDepth 0.42  $\rightarrow$  0.116, surpassing every fixed arm, the supervised baseline at 0.344, and even RESfM-released at 0.209, and 1DSfM 12.4  $\rightarrow$  5.8 vs. the baseline’s 11.0. The performance is thus already latent in the label-free mechanism family; selection, not mechanism design, is the bottleneck. The signal selection needs is not contamination, and realizing this headroom without ground truth is future work.

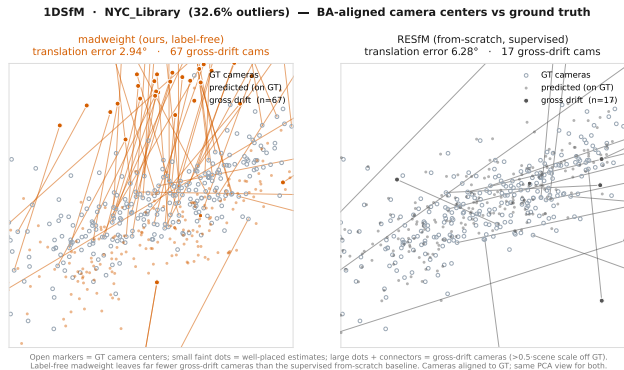


Figure 4. Qualitative 1DSfM (NYC\_Library, 32.6% outliers). BA-aligned camera centers vs. GT (open markers); large dots + connectors mark gross-drift cameras. madweight (left, 2.94°) has fewer and shorter drifts than RESfM-from-scratch (right, 6.28°).

## 5.8. Qualitative

Fig. 4 overlays BA-aligned camera centers on ground truth for 1DSfM NYC\_Library. Label-free madweight (2.94°) leaves far fewer gross-drift cameras than the supervised from-scratch baseline (6.28°), which shows more and larger gross drifts.

## 5.9. Preliminary: training-pool diversity and the label ceiling

A preliminary experiment (single seed, our earlier five-milestone schedule, not the main-table recipe) retrains both methods on progressively diversified pools: MegaDepth-27  $\rightarrow$  +ETH3D (39 scenes, added data  $\sim$ 8% contaminated)  $\rightarrow$  +VGG/T&T (52 scenes, adding a  $\sim$ 16%-contaminated mid-band). The supervised baseline’s clean-OOD accuracy degrades monotonically with diversity (Strecha 0.006  $\rightarrow$  0.77  $\rightarrow$  2.83); in distribution the same diversification is nearly free (MegaDepth 0.33  $\rightarrow$  0.32  $\rightarrow$  0.44, the first step within noise), so the collapse is specific to clean OOD generalization, not a general capacity cost. Meanwhile the clean addition yielded no high-contamination gain and only the contaminated mid-band coincided with one (1DSfM 11.0/11.5/8.8; differences within single-seed noise on this benchmark, so consistent with, though not established by, these runs). If it holds, the implication is pointed: the data that helps is precisely the data whose COLMAP labels degrade (cf. the label ceiling), making supervised diversity scaling self-limiting. The label-free arms diverge by mechanism: soft weighting collapses on clean OOD like the baseline (BlendedMVS 2.9–4.0), whereas madweight is diversity-robust (BlendedMVS 0.22–0.30) and reaches our best high-contamination result anywhere (1DSfM

6.4 at 52 scenes). We report this as a labeled preliminary observation; a canonical-schedule, multi-seed replication is in progress.

## 6. Limitations

(1) No single deployable configuration wins all datasets; selecting the best mechanism per dataset (Table 1) requires knowing the contamination level. We present a study of the phenomenon, not a turnkey adaptive method; an unsupervised selector is future work. (2) Supervised RESfM retains in-distribution MegaDepth (the self-supervised arms come within  $\sim$ 15–20%, no closer) and the clean Strecha mean, and RESfM-released remains stronger than either from-scratch arm on MegaDepth, so we claim no SOTA. (3) Versus supervision, madweight has no stable winner at moderate or extreme ( $17.0 \pm 4.4$  vs. 16.2) contamination; plain remove leads on every evaluated seed at both 43% and  $\sim$ 60%, but against a single-seed baseline (its replicates were still training at submission), and the fully seed-robust claims remain the mechanism ordering and the clean-OOD wins. (4) OOD tracks are self-constructed; OOD numbers are controlled within our pipeline and not comparable to RESfM’s published OOD numbers. (5) Two symmetric-structure scenes (Ellis Island, Tower of London) fail for all methods and inflate 1DSfM means (we report medians alongside); these are the cases where excess removal can disconnect the viewing graph [1].

## 7. Conclusion

SS-RESfM shows that deep SfM outlier handling can be made entirely self-supervised (no ground-truth or COLMAP labels) while staying within  $\sim$ 15–20% of the supervised baseline in distribution, winning the clean-OOD splits, and matching it at extreme contamination, where both beat COLMAP and supervision itself can no longer be manufactured. The optimal mechanism is contamination-dependent: removal at high contamination, reweighting at low. Robust deep SfM should treat outlier handling as adaptive and self-supervised rather than a single supervised mechanism. Code, configurations, and the rebuilt OOD tracks will be made publicly available upon publication.

## References

- [1] Khatib, Kasten, Moran, Galun, and Basri. RESfM: Robust Deep Equivariant Structure from Motion. ICLR 2025. 1, 2, 3, 6, 8
- [2] Moran, Koslowsky, Kasten, Maron, Galun, and Basri. Deep Permutation Equivariant Structure from Motion. ICCV 2021. 1, 2

- 585 [3] Brynte, Iglesias, Olsson, and Kahl. Learning Structure-  
586 from-Motion with Graph Attention Networks. CVPR  
587 2024. 2
- 588 [4] Schönberger and Frahm. Structure-from-Motion Revis-  
589 ited. CVPR 2016. 1
- 590 [5] Rousseeuw and Croux. Alternatives to the Median Ab-  
591 solute Deviation. JASA 1993. 2
- 592 [6] Wilson and Snavely. Robust Global Translations with  
593 1DSfM. ECCV 2014. 3
- 594 [7] Li and Snavely. MegaDepth: Learning Single-View  
595 Depth Prediction from Internet Photos. CVPR 2018.  
596 3
- 597 [8] Strecha, von Hansen, Van Gool, Fua, and Thoennessen.  
598 On Benchmarking Camera Calibration and Multi-View  
599 Stereo for High-Resolution Imagery. CVPR 2008. 3
- 600 [9] Yao et al. BlendedMVS: A Large-Scale Dataset for Gen-  
601 eralized Multi-View Stereo Networks. CVPR 2020. 3
- 602 [10] Olsson and Enqvist. Stable Structure from Motion for  
603 Unordered Image Collections. SCIA 2011. 3